

Joonas Tuhkuri

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Current Positions

Assistant Professor of Economics, Stockholm University	2023–
Senior Fellow, Etla	
Research Affiliate, RFBerlin, CESifo, IZA	

Education

PhD in Economics, MIT	2022
MS in Economics, University of Helsinki	2015
BS in Economics, University of Helsinki	2013
Graduate Student Affiliate, Harvard University	2019–22
Visiting Graduate Student, University of Toronto	2014

Selected Awards

Upjohn Institute Dissertation Award (first prize)	2022
Harvard CES Dissertation Completion Fellowship	2021
35 under 35, HS Visio	2021
Hausman Fellowship, MIT	2020
35 under 35, Kauppalehti	2019
MIT Center for International Studies Grant	2019–20
Kone Foundation Grant	2018
Labor Foundation Grant	2018–20
Stanley and Rhoda Fischer Fellowship, MIT	2017
Emil Aaltonen Foundation Grant	2017–21
Castle Krob International Fellowship, MIT	2017
Kordelin Foundation Grant	2016–20
Yrjö Jahnsson Foundation Grant	2016–20
Fulbright Fellowship	2015
Best Economics Thesis Award, University of Helsinki	2015
Best Economics Student, Väinö Linna Award, University of Helsinki	2015
Best Thesis, ARC Award, The US Embassy in Finland	2015

Research Funding

Handelsbankens forskningsstiftelser (SEK 300 000)	2025
OP Group Research Foundation (€7000)	2025
OP Group Research Foundation (\$54k)	2020–21
Ministry of Economic Affairs and Employment (\$21k)	2020
Yrjö Jahnsson Foundation (\$22k)	2020
Foundation for Economic Education (\$20k)	2020
George and Obie Shultz Fund (\$17k)	2019–20

Working Papers

What Do Technology Grants Do?

(with Johannes Hirvonen and Aapo Stenhammar)

conditionally accepted at the Quarterly Journal of Economics

We present new evidence on the impact of EU technology grants on employment and skill mix in Finnish small and medium-sized manufacturing firms, 1994–2018. The subsidies funded new machinery, including robots. Comparing close grant winners and losers, we find that subsidized investments raised employment by 23 percent with no detectable change in the skill mix. We use machine learning on the text of application evaluations to match firms, and we analyze firms' stated plans and later outcomes. Our evidence suggests that these grants more often supported expansion—new products and markets—than directly automating work. By contrast, outside the program, IT investment is more strongly associated with skill upgrading than machinery investment. In this context, the technology grants raised employment while leaving the skill mix unchanged. The jobs created were primarily held by non-college workers.

The Economist: Economists are revising their views on robots and jobs, 2022.

The Economist: Leader: The world should welcome the rise of the robots, 2022.

Wired: Automation Isn't the Biggest Threat to US Factory Jobs, 2022.

Evolving Returns to Personality

(with Ramin Izadi)

accepted at the Journal of Labor Economics

We use cognitive and personality test records of half a million Finnish men to describe the trends in labor-market returns to psychological traits between 2001 and 2015. Consistent with earlier research, we observe a decline in the value of cognitive skills and an increase in the value of noncognitive skills. Our novel results show that extraversion-related traits drive the rising returns to noncognitive skills. The returns to conscientiousness-related traits have remained stable. The increasing gains from extraversion are driven by those at the lower end of earnings and by the employment margin. We also show that these traits predict different paths in the labor market. Extraversion predicts lower education and more work experience. Cognitive ability and conscientiousness are linked to sorting into highly educated and high-paying jobs.

Psychological Traits and Adaptation in the Labor Market

(with Ramin Izadi)

Labor markets are in constant change. Which personality traits and skills help workers to deal with a changing environment? This paper documents how responses to labor-market shocks vary by individuals' psychological traits. We construct measures of cognitive ability, extraversion, and conscientiousness using standardized personality and cognitive tests administered during military service to approximately 80% of Finnish men born 1962–1979. We analyze establishment closures and mass layoffs between 1995–2010 and document heterogeneous responses to the shock. Extraversion is the strongest predictor of adaptation: the negative effect of a mass layoff on earnings is about 20% smaller for those with one standard deviation higher scores of extraversion. Conscientiousness appears to have no differential impact conditional on other traits. Cognitive ability and education predict a significantly smaller initial drop in earnings but have no long-term advantage. Our findings appear to be driven directly by smaller dis-employment effects: extraverted and high cognitive-ability individuals find re-employment faster in a similar

occupation and industry they worked in before. Extraversion's adaptive value is robust to controlling for pre-shock education, occupation, and industry, which rules out selection into different careers as the driving mechanism. Extraverts are slightly more likely to retain employment in their current establishment during a mass layoff event, but the retention effect is not large enough to explain the smaller earnings drop.

The Surprising Intergenerational Effects of Manufacturing Decline

This paper analyzes the impact of manufacturing decline on children. To do so, it considers local employment structure—characterizing lost manufacturing jobs and left-behind places—high-school dropout rates, and college access in the US over 1990–2010. To establish a basis for causal inference, the paper uses variations in trade exposure from China, following its entry to the WTO, as an instrument for manufacturing decline in the US. While the literature on job loss has emphasized negative effects on children, the main conclusion of this research is that the rapid US manufacturing decline decreased high-school dropout rates and possibly increased college access. The magnitudes of the estimates suggest that for every 3-percentage-point decline in manufacturing as a share of total employment, the high-school dropout rate declined by 1 percentage point. The effects are largest in the areas with high racial and socioeconomic segregation and in those with larger African American populations. The results are consistent with the idea that the manufacturing decline increased returns and decreased opportunity costs of education, and with sociological accounts linking working-class environment and children's education.

Work in Progress

Scarcity vs. Surplus: New Evidence on Technology and Labor Supply

(with Jonas Mueller-Gastell)

Does shortage of labor or abundance of labor encourage technology adoption? Are machines and men substitutes so that labor scarcity induces investment in technology, or are they complements so that availability of workers facilitates technology adoption? The project uses local labor supply shocks in Finland at the verge of industrialization to study how technology and labor supply interact. These shocks come from two sources: combat deaths and evacuations from invaded areas into designated towns during the Second World War, 1939–45. The project uses newly digitized local and plant-level data on technology use by type, employment, and organization. We find a positive effect of labor abundance on manufacturing development. Evidence on horsepower per person shows that additional labor does not crowd out capital but complements capital investment. Manufacturing employment share and gross value-added per person increase substantially across all identification strategies, including strategies based on military and evacuation plans.

Work Experience

Postdoctoral Fellow, Rockwool Foundation Berlin and the Berlin School of Economics	2022–23
Research Assistant, MIT, RA to Daron Acemoglu, David Autor, and John Van Reenen	2017–19
Visiting Researcher, Aalto University	2017–22
Visiting Researcher, Etla Economic Research	2017–22
Researcher, Etla Economic Research	2014–16
Research Internship, Ministry of Finance of Finland	2013

Teaching Experience

Labor Economics, Stockholm University	2024–
Economics and Public Policy, Stockholm University	2024–
PhD Labor Economics, Stockholm University	2023
PhD Labor Economics, MIT, TA to David Autor and Arindrajit Dube	2020
Introduction to Economics, University of Helsinki, TA	2012–14
Keynotes and Executive Education Workshops	2015–

Publicly Available Data

ETLAnow: Real-time unemployment forecasts based on Google search data. 2014–2024.
Featured in The Washington Post, Bloomberg, Chicago Tribune, YLE, and widely in the global media.
[Bloomberg](#): *Understanding Europe's Economy in 100 Billion Google Searches*, 2016.

[Occupation Codes](#): This package harmonizes Finnish occupation codes.

[Industry Codes](#): This package harmonizes Finnish industry codes (2 and 3-digit levels).

[County and Subregion Codes](#): This package harmonizes Finnish county codes into consistent regions.

Selected Presentations

2025: NBER Analysis of Business Taxation, Princeton, Federal Reserve Bank of Chicago, Workshop on Skills, Tasks & Technologies in the AI Era, WEI 2025: Wages, Employment and Inequality, Workshop Future Inequalities: Boys and Men in Focus, TELA, Ifo Institute for Economic Research

2024: Conference on Technological Change at UCLouvain, IIES, IFAU, SOFI, Nordic Public Policy Symposium at Copenhagen University, Yhteiskuntatieteilijäpäivät

2023: ASSA, Sciences Po/Paris School of Economics, University of Oslo, European Labor Symposium for Early Career Economists, Nuremberg Research Seminar in Economics, CReAM–RFBerlin workshop, Copenhagen Business School.

2022: NBER Summer Institute, Columbia Business School, University of Wisconsin-Madison, ETH Zurich, Econometric Society, Stockholm University, Uppsala University, LMU Munich, Berlin School of Economics, Humboldt University of Berlin, BI Oslo, IFN, MIT Industrial Performance Center, CReAM, CESifo.

pre–21: MIT, Harvard, Boston University, European Parliament, European Commission, Helsinki GSE, Aalto University, Etla Economic Research, Labore, University of Helsinki, University of Jyväskylä, Ministry of Finance of Finland, Ministry of Economic Affairs and Employment of Finland.

Reports

Policy Brief: New Evidence on the Effect of Technology on Employment and Skill Demand, ETLA Brief 108, 2022 (with J. Hirvonen and A. Stenhammar).

Forecasting Unemployment with Google Searches, ETLA Working Paper 35, 2016.

ETLAnow: A Model for Forecasting with Big Data, ETLA Report 54, 2016.

Big Data: Do Google Searches Predict Unemployment?, University of Helsinki, 2015.

Big Data: Google Searches Predict Unemployment in Finland, ETLA Report 31, 2014.

International Sourcing in Finland and Sweden, ETLA B 275, 2017 (with Hans Lööf et al.).

Globalization Threatens One Quarter of Finnish Employment, ETLA Brief 46, 2016.

Finland in Global Value Chains, Prime Minister's Office 11/2016, 2016 (with J. Ali-Yrkkö et al.)

Offshoring R&D, CESIS Working Paper 439, 2016 (with H. Lööf, A. Mohammadi, and P. Rouvinen).

Trade and Innovation: Matched Worker-Firm-Level Evidence, ETLA Working Paper 39, 2016.

Women and Men in Central Government 2012, Ministry of Finance, 20/2013, 2013.

Professional Service

Referee service: American Economic Review, Quarterly Journal of Economics, Journal of the European Economic Association, Economic Journal, Journal of Public Economics, Journal of Labor Economics, Labour Economics, Journal of Applied Econometrics, Journal of Human Resources, AER Insights.

Department service: Search Committee at Stockholm University 2024, Handbook of Labor Economics Conference Committee 2023, Berlin School of Economics PhD Recruiting Committee 2023, RF Berlin Hiring Committee 2022, Labor Coffee at MIT, 2020–22.

Basic Info

Fields of Specialization: Labor Economics, Technology and Innovation, Psychology and Economics.

Languages: English (fluent), Finnish (native), Swedish (basic)

Military service: Finnish Defence Forces, 2010.

Citizenship: Finland (EU)

References

Professor **David Autor**, MIT Department of Economics, dautor@mit.edu.

Professor **Daron Acemoglu**, MIT Department of Economics, daron@mit.edu.

Professor **Simon Jäger**, Princeton Department of Economics, sjaeger@mit.edu.

Updated January 2026.